

## CMPG222 – Practical Assignment 4

Dear students

Please do the following assignments at the end of **Chapter 6: 19, 20, 23, and 24**. The LOs (learning objectives) addressed by the assignments are indicated as part of the assignment. To complete the tasks, you must study Sections 6.1 to 6.2 in the textbook. Please note that the data files you must use to do the assignments are shown to the left of the work in the book. These files are available on eFundi under the Resources tool. Copy the relevant data to your sheet and use the data to answer the assignment. Submit the answers to these assignments as **a single Microsoft Excel document** on the eFundi assignment. Create a separate sheet for each of the four questions like the following example:

|             |             |             |                    |
|-------------|-------------|-------------|--------------------|
| Question 19 | Question 20 | Question 23 | <u>Question 24</u> |
|-------------|-------------|-------------|--------------------|

Also, remember to show the sub-sections of each question clearly in each sheet, e.g.:

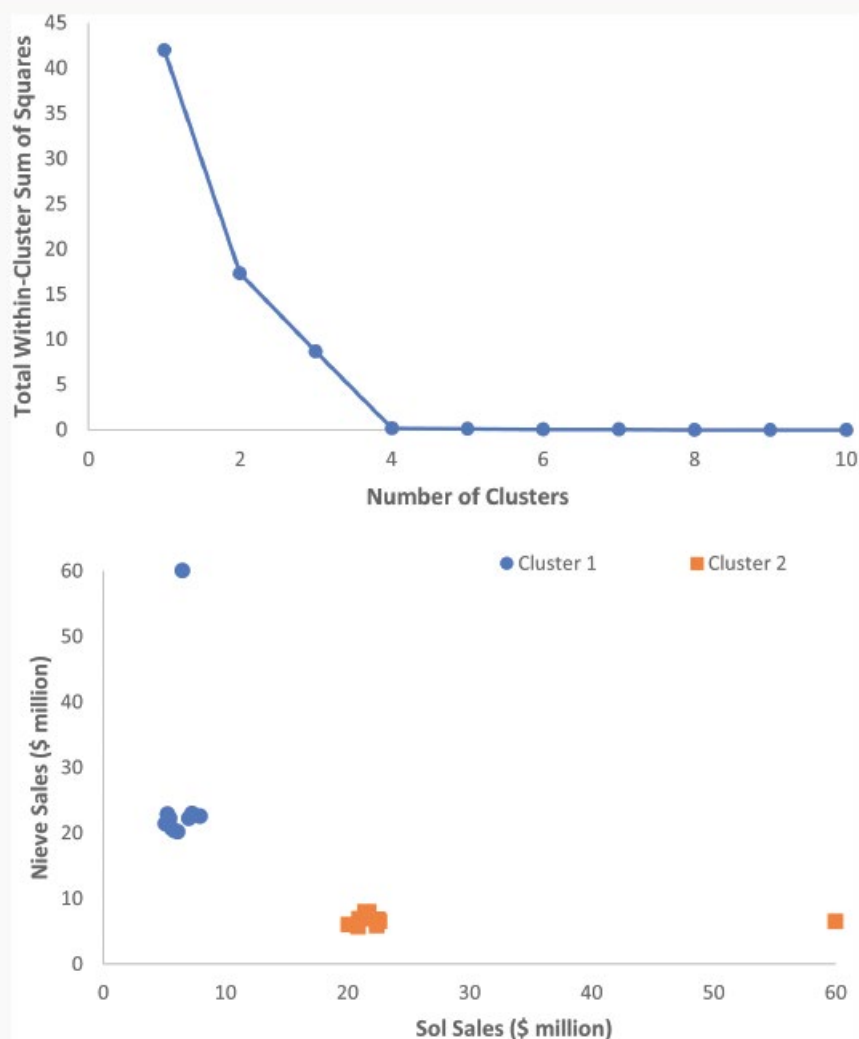
|                                                  |  |  |  |  |
|--------------------------------------------------|--|--|--|--|
| Question 17 (a)                                  |  |  |  |  |
| The answer to Question 17(a) should appear here. |  |  |  |  |
| Question 17(b)                                   |  |  |  |  |
| The answer to Question 17(b) should appear here. |  |  |  |  |

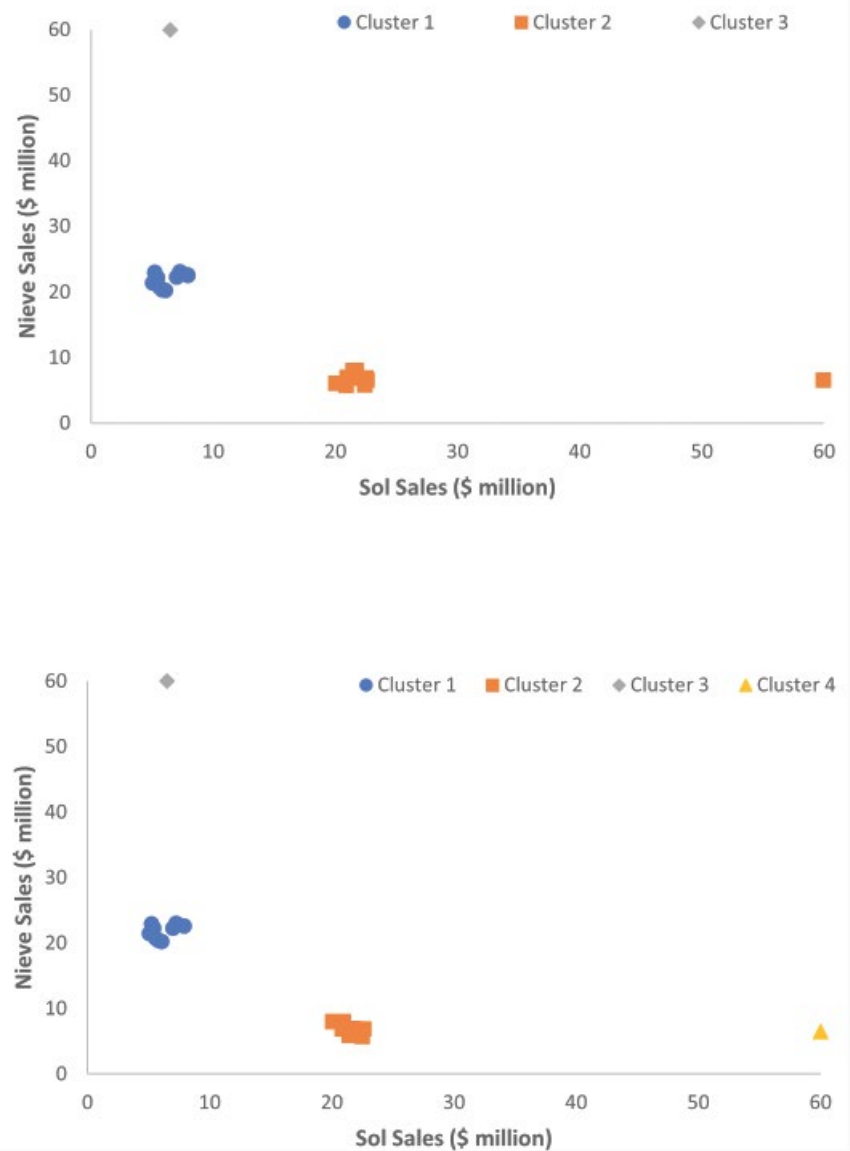
Kind regards

The CMPG222 lecturers

**19. Outliers' Impact on Clustering.** Sol & Nieve is a sporting good and outdoor gear retailer that operates in North America and Central America. In an attempt to characterize its stores and reassess Sol & Nieve's supply chain operations, Gustavo Esposito is analyzing sales in 22 regions for its two primary product lines: sol (beach-oriented apparel) and nieve (mountain-oriented apparel). For a range of  $k$  values, Gustavo has generated the following table of average silhouette values and elbow chart. After examining this output, Gustavo also generated the following scatter charts for the clusters correspond to  $k = 2, 3$ , and 4. [LO 4](#)

| $k$ | Average Silhouette Value |
|-----|--------------------------|
| 2   | 0.741                    |
| 3   | 0.779                    |
| 4   | 0.837                    |
| 5   | 0.667                    |
| 6   | 0.401                    |
| 7   | 0.433                    |
| 8   | 0.502                    |
| 9   | 0.536                    |
| 10  | 0.494                    |

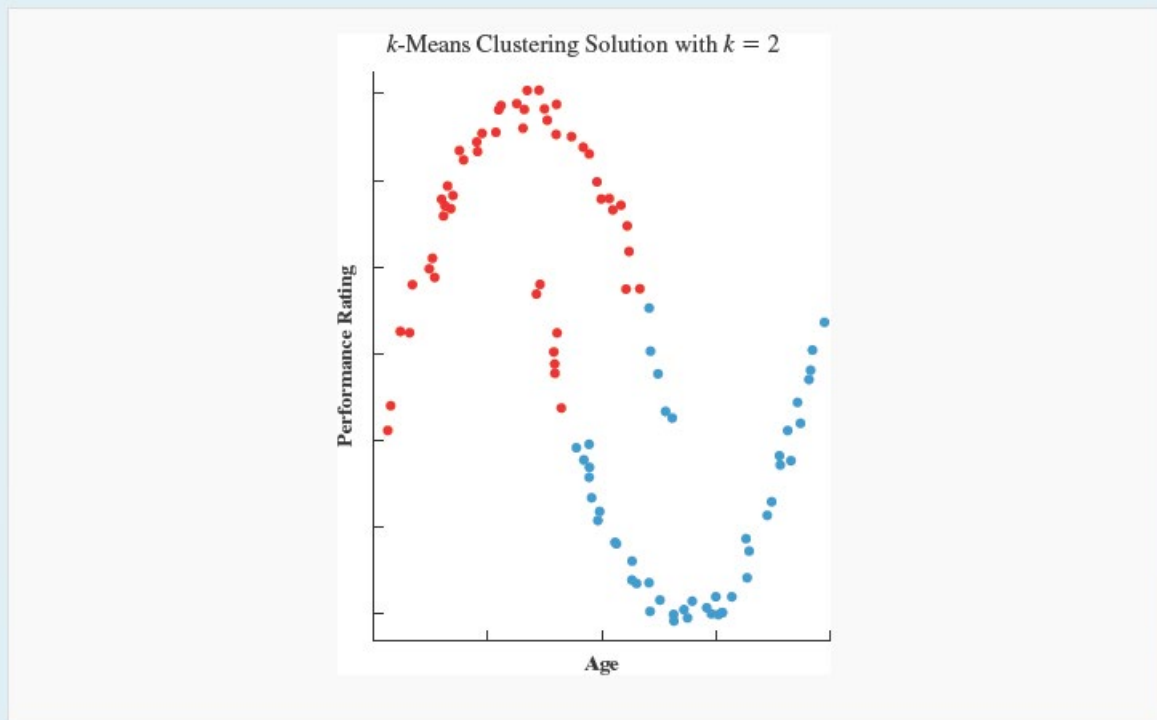




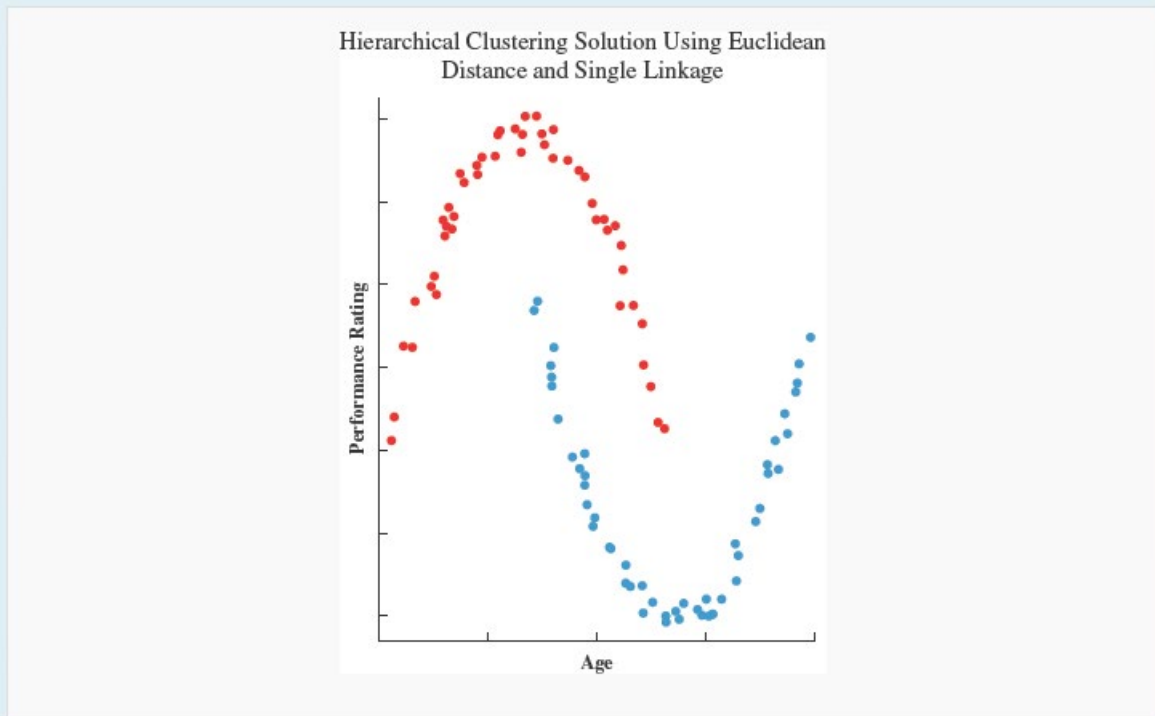
- What value of  $k$  is the most appropriate for these data?
- For the value of  $k$  identified in part (a), how should Gustavo interpret the results to characterize the sales regions?

20. **Cluster Shapes for  $k$ -Means versus Single Linkage.** Heidi Zahn is a human resources manager currently reviewing data on 98 employees. In the data, each observation consists of an employee's age and an employee's performance rating. LO 4 , 5 , 6

- a. Heidi applied  $k$ -means clustering with  $k = 2$  to the data and generated the following plot to visualize the clusters. Based on this plot, qualitatively characterize the two clusters of employees categorized by the  $k$ -means approach.



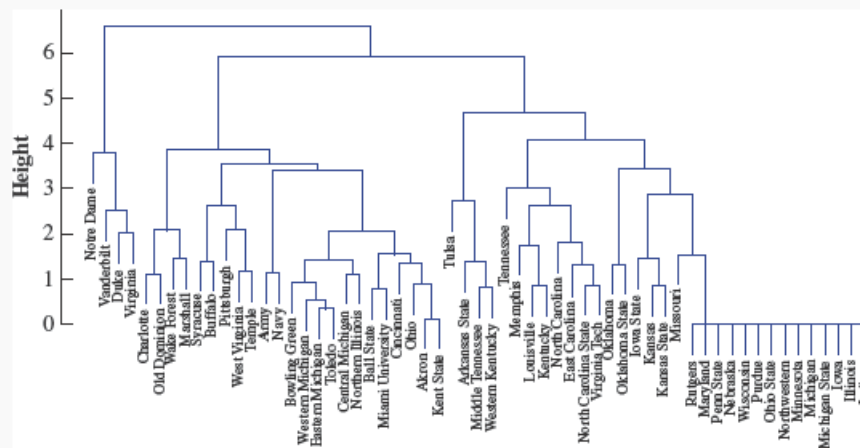
- b. For a comparison, Heidi applied hierarchical clustering with the Euclidean distance and single linkage to the data and generated the following plot based on the level of agglomeration with two clusters. Based on this plot, qualitatively characterize the two clusters of employees categorized by the hierarchical approach.



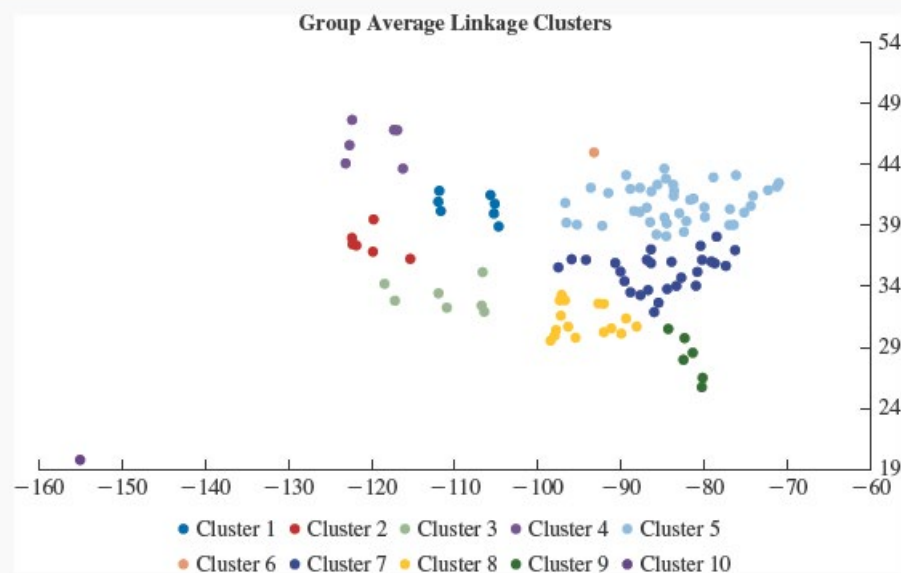
- c. Which of the two approaches (*k*-means clustering from part (a) or hierarchical clustering from part (b)) would you recommend?

**23. Interpreting Merge Steps from a Dendrogram.** From 1946 to 1990, the Big Ten Conference consisted of the University of Illinois, Indiana University, University of Iowa, University of Michigan, Michigan State University, University of Minnesota, Northwestern University, Ohio State University, Purdue University, and University of Wisconsin. In 1990, the conference added Pennsylvania State University. In 2011, the conference added the University of Nebraska. In 2014, the University of Maryland and Rutgers University were added to the conference with speculation of more schools being added in the future.

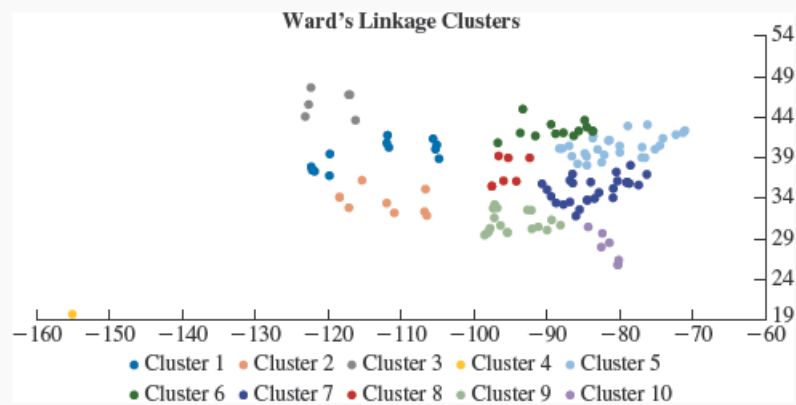
Based on the football stadium capacity, latitude, longitude, and enrollment, the Big Ten commissioner is curious how a clustering algorithm would suggest the conference expand beyond its current 14 members. To represent the 14 member schools as an entity, each variable value for the 14 schools of the Big Ten Conference has been replaced with the respective variable median over these 14 schools. Using Euclidean distance to measure dissimilarity between observations, hierarchical clustering with complete linkage generates the following dendrogram. Describe the next three stages of conference expansion plan suggested by the dendrogram. [LO 6](#)



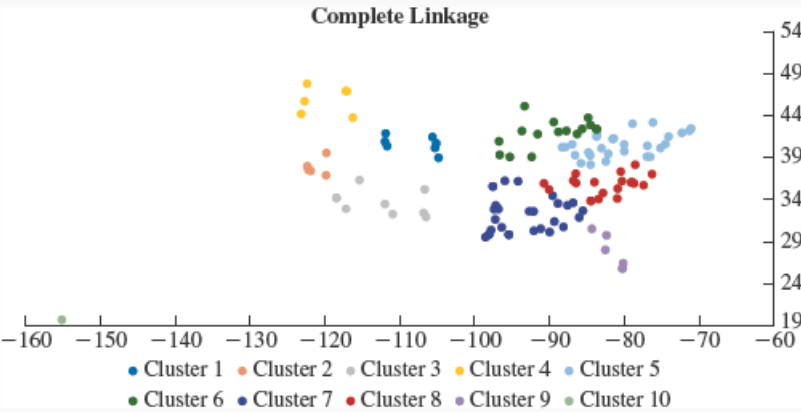
**24. Comparing Different Linkage Methods.** The Football Bowl Subdivision (FBS) level of the National Collegiate Athletic Association (NCAA) consists of over 100 schools. Most of these schools belong to one of several conferences, or collections of schools, that compete with each other on a regular basis in collegiate sports. Suppose the NCAA has commissioned a study that will propose the formation of conferences based on the similarities of constituent schools. If the NCAA cares only about geographic proximity of schools when determining conferences, it could use hierarchical clustering based on the schools' latitude and longitude values and Euclidean distance to compute dissimilarity between observations. The following charts and tables illustrate the 10-cluster solutions when using various linkage methods (group average, Ward's, complete, and single) to determine clusters of schools to be assigned to conferences using hierarchical clustering. Compare and contrast the resulting clusters created using each of these different linkage methods for hierarchical clustering. [LO 5](#) , [6](#)



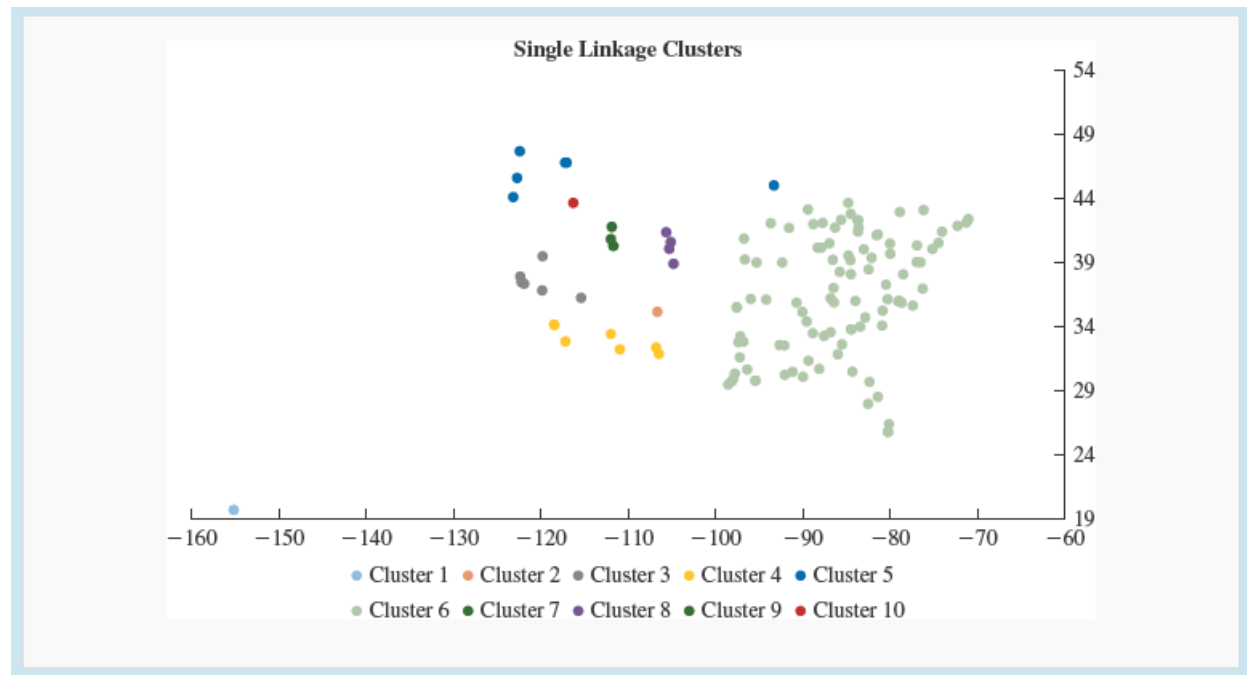
| Group       | Count  | Min of   | Max of   | Min of    | Max of    |
|-------------|--------|----------|----------|-----------|-----------|
| Average     | of     | Latitude | Latitude | Longitude | Longitude |
| Linkage     | School |          |          |           |           |
| Clusters    |        |          |          |           |           |
| 1           | 7      | 38.9     | 41.7     | -111.9    | -104.8    |
| 2           | 6      | 36.2     | 39.4     | -122.3    | -115.3    |
| 3           | 8      | 31.8     | 35.1     | -118.4    | -106.4    |
| 4           | 6      | 43.6     | 47.6     | -123.1    | -116.2    |
| 5           | 43     | 38.0     | 43.6     | -96.7     | -71.0     |
| 6           | 1      | 45.0     | 45.0     | -93.3     | -93.3     |
| 7           | 30     | 31.8     | 38.0     | -97.5     | -76.2     |
| 8           | 18     | 29.5     | 33.2     | -98.5     | -88.1     |
| 9           | 7      | 25.8     | 30.5     | -84.3     | -80.1     |
| 10          | 1      | 19.7     | 19.7     | -155.1    | -155.1    |
| Grand Total | 127    | 19.7     | 47.6     | -155.1    | -71.0     |



| Ward's Linkage Clusters | Count of School | Min of Latitude | Max of Latitude | Min of Longitude | Max of Longitude |
|-------------------------|-----------------|-----------------|-----------------|------------------|------------------|
| 1                       | 12              | 36.8            | 41.7            | 122.3            | 104.8            |
| 2                       | 9               | 31.8            | 36.2            | -118.4           | -106.4           |
| 3                       | 6               | 43.6            | 47.6            | -123.1           | -116.2           |
| 4                       | 1               | 19.7            | 19.7            | -155.1           | -155.1           |
| 5                       | 28              | 38.0            | 43.0            | -88.3            | -71.0            |
| 6                       | 13              | 10.8            | 45.0            | -96.7            | -83.6            |
| 7                       | 20              | 31.8            | 38.0            | -90.7            | -70.2            |
| 8                       | 7               | 35.5            | 39.2            | -97.5            | -92.3            |
| 9                       | 18              | 29.5            | 33.2            | -98.5            | -88.1            |
| 10                      | 7               | 25.8            | 30.5            | -84.3            | -80.1            |
| Grand Total             | 127             | 19.7            | 47.6            | -155.1           | -71.0            |



| Complete Linkage Clusters | Count of School | Min of Latitude | Max of Latitude | Min of Longitude | Max of Longitude |
|---------------------------|-----------------|-----------------|-----------------|------------------|------------------|
| 1                         | 7               | 38.9            | 41.7            | -111.9           | -104.8           |
| 2                         | 5               | 36.8            | 39.2            | -122.3           | -119.7           |
| 3                         | 9               | 31.8            | 36.2            | -118.4           | -106.4           |
| 4                         | 6               | 43.6            | 47.6            | -123.1           | -116.2           |
| 5                         | 28              | 38.0            | 43.0            | -88.3            | -71.0            |
| 6                         | 16              | 39.0            | 45.0            | -96.7            | -83.6            |
| 7                         | 28              | 29.5            | 36.1            | -98.5            | -85.5            |
| 8                         | 20              | 33.8            | 38.0            | -90.7            | -76.2            |
| 9                         | 7               | 25.8            | 30.5            | -84.3            | -80.1            |
| 10                        | 1               | 19.7            | 19.7            | -155.1           | -155.1           |
| Grand Total               | 127             | 19.7            | 47.6            | -155.1           | -71.0            |



| Single Linkage Clusters | Count of School | Min of Latitude | Max of Latitude | Min of Longitude | Max of Longitude |
|-------------------------|-----------------|-----------------|-----------------|------------------|------------------|
| 1                       | 1               | 19.7            | 19.7            | -155.1           | -155.1           |
| 2                       | 1               | 35.1            | 35.1            | -106.6           | -106.6           |
| 3                       | 6               | 36.2            | 39.4            | -122.3           | -115.3           |
| 4                       | 7               | 31.8            | 34.1            | -118.4           | -106.4           |
| 5                       | 1               | 45.0            | 45.0            | -93.3            | -93.3            |
| 6                       | 98              | 25.8            | 43.6            | -98.5            | -71.0            |
| 7                       | 5               | 44.1            | 47.6            | -123.1           | -117.0           |
| 8                       | 1               | 43.6            | 43.6            | -116.2           | -116.2           |
| 9                       | 3               | 40.2            | 41.7            | -111.9           | -111.6           |
| 10                      | 4               | 38.9            | 41.3            | -105.6           | -104.8           |
| Grand Total             | 127             | 19.7            | 47.6            | -155.1           | -71.0            |